

Original article

Manual or exercise therapy for long-standing adductor-related groin pain: A randomised controlled clinical trial

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ARTICLE INFO

Article history:

Received 6 November 2009

Received in revised form

13 July 2010

Accepted 14 September 2010

Keywords:

Groin injury
osteitis pubis
adductor tendonitis
adductor injury

ABSTRACT

Hypothesis: A multi-modal treatment program (MMT) is more effective than exercise therapy (ET) for the treatment of long-standing adductor-related groin pain.**Study design:** Single blinded, prospective, randomised controlled trial.**Methods:** *Patients:* Athletes with pain at the proximal insertion of the adductor muscles on palpation and resisted adduction for at least two months. *Interventions:* ET: a home-based ET and a structured return to running program with instruction on three occasions from a sports physical therapist. MMT: Heat, Van den Akker manual therapy followed by stretching and a return to running program. *Primary outcome:* time to return to full sports participation. Secondary outcome measures: objective outcome score and the visual analogue pain score during sports activities. Outcome was assessed at 0, 6, 16 and 24 weeks.**Results:** Athletes who received MMT returned to sports quicker (12.8 weeks, SD 6.0) than athletes in the ET group (17.3 weeks, SD 4.4, $p = 0.043$). Only 50–55% of athletes in both groups made a full return to sports. There was no difference between the groups in objective outcome ($p = 0.72$) or VAS during sports ($p = 0.12$).**Conclusions:** The multi-modal program resulted in a significantly quicker return to sports than ET plus return to running but neither treatment was very effective.

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1. Introduction

Groin pain is a frequent complaint in athletes. It occurs commonly in sports involving repeated sprinting, twisting, kicking and cutting such as soccer, rugby, Australian rules football and (ice-) hockey (Bradshaw and McCrory, 1997; Jansen et al., 2008b). In football groin injuries have been reported to account for about 10% of all injuries (Hawkins et al., 2001). Most of these injuries will recover quickly and in one study only three of 22 groin injured athletes still had complaints after three weeks (Arnason et al., 2004).

When groin pain does go on to be long-standing it can be hard to treat, with a relatively long period for return to full sports activity (Hölmich et al., 1999). Adductor-related groin pain has been reported to account for 58% of groin injuries in all sports and 69% of groin injuries in footballers (Hölmich, 2007).

Exercise therapy (ET) is recommended as the first line of treatment after a period of rest or restricted activity (Jansen et al., 2008a). In a recent systematic review (Machotka et al., 2009) on the effect of ET only one randomised controlled trial (Hölmich et al., 1999) was identified. Both reviews noted the lack of randomised controlled studies on the treatment of athletic groin injuries and recommended that more trials be performed in this area. **The study of Hölmich et al. showed the value of ET (level 1 evidence) for athletes with long-standing adductor-related groin pain.** 79% of athletes who underwent ET were able to resume sports at their pre-injury level. The median time to return to sport was 18.5 weeks (range 13–26). At present ET can be considered the therapy with the highest level of evidence (usual care) for the treatment of adductor-related groin pain.

The results of a multi-modal treatment program (MMT) using heat, Van Den Akker manual therapy method, stretching and a return to running program have been studied previously retrospectively (Weir et al., 2008). The study reported promising results with 27 of the 30 athletes (90%) being able to return to sporting activities after treatment. A weakness of the study was the retrospective design and the lack of a control group. The aim of the study

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Table 1
Inclusion and exclusion criteria.

Inclusion criteria	Exclusion criteria
Age 18–50 years	Palpable inguinal or femoral hernia or pain felt above the conjoint tendon
Groin pain for at least two months	Clinical signs or symptoms of prostatitis or urinary tract infection
Groin pain during or after sporting activities	Back pain felt from T10 to L5
Desire to return to active sports participation at pre-injury level	Osteoarthritis of the hip
Pain located at the proximal insertion of the adductor muscles on the pubic bone	Clinical suspicion of a nerve entrapment syndrome
Pain felt at the proximal insertion of the adductor muscles on the pubic bone when performing resisted adduction	Inability to perform the active physical training program
	Use of anti-coagulant medication
	Instability of the medial collateral ligament of the ipsilateral knee

Abbreviations: T = thoracic; L = lumbar.

was to compare the new therapy (MMT) to the current therapy with the highest level of evidence (ET (usual care)), for the treatment of long-standing adductor-related groin pain, in a single blinded prospective randomised clinical trial.

2. Methods

2.1. Subjects

Athletes were referred to the sports medicine department of a large district hospital by local physiotherapists, general practitioners and sports medicine physicians. All athletes were assessed for suitability to participate by a single sports physician, experienced in the field of athletic groin injuries, who used a structured

examination protocol. The inclusion and exclusion criteria are shown in Table 1. Physical examination was used to assess for the presence of adductor pain and has been shown to be reliable (Hölmich et al., 2004). All athletes who attended the department were included if they met all the inclusion criteria and consented to participate.

2.2. Design

The study was a single blinded randomised controlled clinical trial. The local medical ethics committee gave permission for the study (METC-Zuid West Holland 05-109).

After informed consent and inclusion the athletes were randomised using sealed envelopes. The athlete chose one of 100 opaque envelopes in the presence of the department's secretary. Appointments for starting the allocated treatment were arranged and treatment was started within one week of randomisation. The examining physician was not involved in the randomisation process and remained unaware of the treatment allocation. Athletes were instructed not to reveal their treatment allocation to the physician at follow up appointments.

2.3. Treatment

2.3.1. Exercise therapy

Athletes were seen by one of two sports physical therapists in the hospital. The physical therapists both received training to ensure a standardised treatment. No treatment other than the ET was given. The athletes were instructed to perform the ET three times a week at home. The ET is described in detail in Table 2 and the exercises are the same as in the study of Hölmich et al. (1999).

Athletes received instructions from the physical therapists on performing the first module on their first visit. After two weeks they attended again and received instructions from the physical therapist on the second home-based module. At six weeks they were seen again and received instructions on the return to running program. During the first six weeks only cycling was allowed. At six weeks the return to running program was started. The return to running program is outlined in Table 3.

Table 2
Exercise therapy.

Module 1–1st two weeks		Module 2– from 3rd week	
Exercise	Amount	Exercise	Amount (all performed twice)
Static adduction against soccer ball placed between feet lying supine.	10 repetitions of 30 s	Leg abduction and adduction exercises performed in side lying	5 series of 10 repetitions of each exercise
Static adduction against soccer ball placed between knees when lying supine	10 repetitions of 30 s	Low-back extension exercises prone over end of couch	5 series of 10 repetitions
Abdominal sit-ups both in straightforward direction and in oblique direction.	5 series of 10 repetitions	One-leg weight pulling abduction/adduction standing	5 series of 10 repetitions for each leg
Combined abdominal sit-ups and hip flexion, starting from supine position and with soccer ball between knees (folding knife exercise)	5 series of 10 repetitions	Abdominal sit-ups both in straightforward direction and in oblique direction	5 series of 10 repetitions
Balance training on wobble board	5 min	One-leg coordination exercise with flexing and extending knee and swinging arms in same rhythm (cross country skiing on one leg)	5 series of 10 repetitions for each leg
One-foot exercises on sliding board, with parallel feet as well as with 90° angle between feet	5 sets of 1 min continuous work with each leg and in both positions	Training in sideways motion on a mini-skateboard	5 min
		Balance training on wobble board	5 min
		Skating movements on sliding board	5 sets of 1 min continuous work

Table 3

Return to running program. (Hogan, 2006).

Return to running program		
Phase 1 Slow jogging	Phase 2 Straight sprints	Phase 3 Cutting
Alternate day jogging beginning with 5 min and increasing with 5 min per run to a total time of 30 min.	100m sprints. 1st 10m for acceleration and last 10m for deceleration. First 6–8 repetitions at 60% of maximum speed.	Sport specific sprints involving changing directions. Beginning with 6–8 repetitions at 60% of maximum speed.
Slow running at easy pace.	First increase speed and later repetitions. Build up to 15–20 repetitions	First increase speed and later repetitions up to 15 repetitions
Can progress to phase 2 when 30 min of jogging provokes no pain.	Can progress to phase 3 when 15 straight sprints provoke no pain	Can progress to sports when 15 cutting sprints can be performed without provoking pain.

At all times athletes were instructed to stop exercising if they experienced groin pain. The exercises were continued for a minimum of eight weeks. They could be stopped when there was no pain felt during or after performing the exercises or running. Compliance was measured using a self completed training diary which was filled in after each exercise session.

2.3.2. Multi-modal treatment

The MMT comprised heat followed by manual therapy after which stretches were performed and then the return to running program. The manual therapy treatment was performed by one of three sports medicine physicians. Before the manual therapy the adductor muscle group is warmed using paraffin packs for 10 min. The paraffin is heated to 60 °C. This is then wrapped in a towel and placed on the proximal insertion of the adductor muscle while the athlete lies semi-prone. The contra lateral hand is then used to control the tension in the adductor muscles while the ipsilateral hand is used to move the hip from a neutral position into flexion, external rotation and abduction while keeping the knee in extension (See Figs. 1 and 2).

The treating physician controls the tension subjectively and applies the maximum tolerable stretch to the adductor muscles. After the movement has been performed the adductor muscle group is compressed with one hand while the other hand moves

the hip into adduction and slight flexion. This circular motion followed by compressions lasts about 25 s and is repeated three times in one treatment session.

After the treatment the athletes are instructed to perform a 5 min warming-up each day using slow jogging or cycling. After warming-up the athlete performs stretches of the adductors of both legs. A standing and sitting adductor stretch is performed and each stretch is performed statically three times for duration of 30 s. After the stretches the athlete lies in a warm bath for 10 min. After 14 days of stretching if no pain or discomfort was felt the athlete began using the same return to running program as in ET group (See Table 3). If athletes did not improve after one treatment with MMT then the treatment was repeated once more. If after a second treatment there was no improvement the treatment was not repeated again.

2.4. Outcome and follow up

The athletes were assessed by a single experienced blinded physician at baseline, 6 and 16 weeks after the start of treatment. If the athlete was not fully recovered at 16 weeks but still felt improvement a further assessment at 24 weeks was done. If athletes were recovered at 16 weeks they did not attend follow up after this. At these times the standardised history and physical examination was repeated (Hölmich et al., 2004). The Tegner and



Fig. 1. The contra lateral hand holds the adductor muscles to control the tension.



Fig. 2. The ipsilateral hand moves the hip into external rotation and abduction to stretch the adductor muscles.

Lysholm (1985) activity score pre-injury was recorded. The range of motion of the hip joint was measured using a goniometer with the patient lying and the hip and knee flexed to 90°. This method has been shown to be reliable (Boone et al., 1978; Bovens et al., 1990).

In cases where athletes fully resumed their sporting activities in the same sport and at the previous level then the time to return to sports in weeks from commencement of the treatment was recorded. The objective outcome measures for successful treatment were: no pain on palpation of the adductor tendons at the pubic bone insertion, and no pain during resisted adduction; no groin pain during or after sporting activities performed at the same level as prior to the injury; return to the same sport at the same level without groin pain. If all three measures were achieved the objective outcome was considered excellent, if two measures were achieved the outcome was good, if one measure was achieved then outcome was fair and if none were achieved the outcome was poor (Hölmich et al., 1999). Visual analogue scores for maximum pain during sports were recorded on a scale from 0–100.

2.5. Statistical analysis

A double data entry process was used. The statistician was blinded to the treatment outcome until the analysis was performed. All data analysis was performed using SPSS 15.0 (SPSS inc, Chicago, USA). The demographic data is presented with their means and standard deviations (SD). In cases of skewed distribution the median and inter-quartile range (IQR) are presented. The baseline characteristics, VAS for pain and time to return to sports were compared between groups using the independent *T*-test or the non-parametric Mann–Whitney *U* test if the data was not normally distributed. The objective outcome score was dichotomised with excellent and good being considered a success and fair or poor being a failure. The difference between the groups was examined with the Fisher's test. The sample size was calculated using an alpha level of 0.05 and a power of 0.8. The previous published results of Hölmich et al. (1999) showed a mean time to return to sport of 18.5

weeks (SD 7). Clinical experience with the manual therapy treatment estimated the time to return to sports at 12 weeks. A drop out rate of 10% was assumed. Based on these calculations then at least 21 athletes had to be included in both groups to be adequately powered to detect a difference in the time to return to sporting activities. Outcome was assessed using a per-protocol analysis.

3. Results

One hundred athletes were referred for inclusion. Forty six athletes were not suitable for inclusion (hip joint problem 12, primary iliopsoas pain 10, pain above the conjoined tendon 10, current lower back pain 8, unwilling to participate 6). There were 53 males and one female included in the study. Fig. 3 shows the allocation of the athletes in the study.

Thirty seven (69%) played soccer, three (6%) played rugby, three (6%) did distance running, three (6%) played field hockey, two (4%) were speed skaters, two (4%) played squash and there were four athletes who played other sports (tennis, basketball, track-and-field and handball).

The athletes had received many different treatments prior to their inclusion in the study. Only 7 athletes (13%) had undergone no previous treatment. Fifteen had had one treatment (5 physical therapy with massage and stretches, 5 physical therapy with ET, 3 manipulative therapy for back and pelvis, 2 anti-inflammatory medication). Twenty-five had had two prior treatment modalities (physical therapy with massages or exercises (20), manual therapy with manipulations (15), anti-inflammatory medication (13), injections (2)). Seven athletes had undergone three or more previous treatments prior to inclusion. None of the athletes who had undergone previous ET had performed more than 10 sessions prior to the inclusion.

Six (11%) athletes dropped out during the study (three in each group). The baseline data of these six did not differ significantly from the athletes who completed the study. Three athletes dropped out before starting with treatment as they did not want to receive

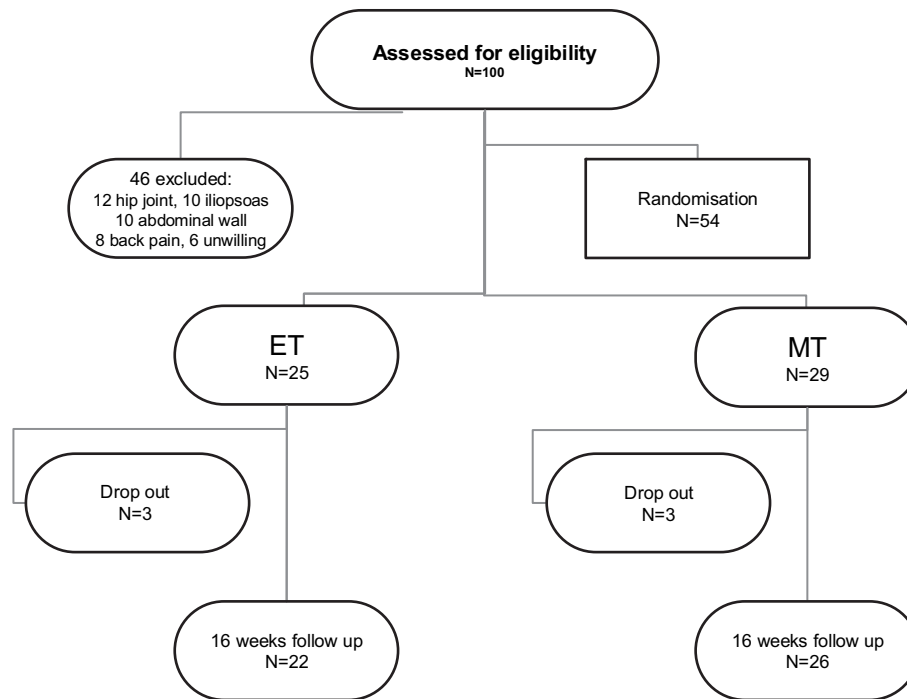


Fig. 3. Flow of athletes through the study.

the allocated treatment, one athlete injured his ankle, one athlete developed lower back pain and decided to change sport and one athlete was lost to follow up. Six athletes in both groups had returned to sports at the 16 weeks follow up appointment. These athletes did not attend follow up at 24 weeks.

The baseline characteristics of the athletes in both groups are shown in the Table 4 below. There were no significant differences between the two groups at baseline ($p > 0.16$).

Table 4
Baseline characteristics of the two groups at baseline.

	Exercise therapy <i>n</i> = 25	Multi-modal <i>n</i> = 29
Age (years)	27.4 (7.3)	28.7 (8.2)
Weight (kilograms)	80.8 (9.7)	82.3 (10.6)
Height (cm)	182 (7)	185 (8)
BMI (kg/m ²)	24.4 (2.5)	24.1 (2.6)
Duration of symptoms (weeks)	32 (IQR 17.5–81.0)	32 (IQR 16.0–72.0)
Absence from sport (weeks)	8 (IQR 4.0–19.0)	14 (IQR 8.0–17.5)
Onset of injury		
Acute	5	9
Gradual	20	20
Location of injury		
Left	7	11
Right	11	12
Bilateral	7	6
VAS pain (0 = no pain, 100 maximum pain) during sports	58.5 (26.2)	58.9 (21.3)
Level of athletic activity prior to injury.		
Elite (>5 times per week.)	6	9
Competitive (3 or 4 times a week)	16	18
Recreational (1 or 2 times a week)	3	2
Sports activities at baseline		
Ceased	18	18
Reduced	7	11
Unchanged	0	0
Tegner activity score pre-injury	8.7 (0.5)	8.7 (0.6)

(No significant differences $p > 0.16$).

3.1. Time to return to sports

Fifty percent (13/26) of athletes treated with the MMT who returned to full sports participation did so in an average of 12.8 weeks (SD 6.0). In the ET group 55% (12/22) returned to full sports participation in 17.3 weeks (SD 4.4). This difference in time to return to sport was statistically significant ($p = 0.043$). The difference in percentage of athletes making a full return to sport was not significant ($p = 0.78$).

3.2. Objective treatment outcome

The objective treatment outcome in the two groups is shown in Table 5. There was no significant difference in objective outcome between the two groups ($p = 0.72$).

Of the athletes treated with MMT 11 received one treatment and 15 received two treatments. There was no correlation between the number of treatments and the outcome ($p = 0.404$). In the ET group the mean number of sessions performed was 75% (IQR 60%–90%). There was no correlation between the compliance and the outcome ($p = 0.318$).

3.3. VAS pain score during sports

The VAS pain scores at 0 and 16 weeks during sports improved significantly in both groups. In the manual therapy group the VAS

Table 5
Distribution of objective treatment outcome in the two groups.

		Treatment	
		ET	MMT
Objective outcome	Poor	6	11
	Fair	4	1
	Good	7	7
	Excellent	5	7
Total		22	26

score during sport improved from 58.9 (SD 21.3) before treatment to 36.1 (SD 30.1) after treatment ($p = 0.01$). In the ET group the VAS during sports improved from 58.5 (SD 26.2) before treatment to 21.0 (SD 27.0) after treatment ($p = 0.000$). The difference in VAS during sport between the two groups was not significant ($p = 0.12$).

The mean hip rotation at baseline was: internal 25° (SD 11), external rotation 39° (SD 12) and total 64° (SD 17). The range of motion of the hip joint did not alter significantly after treatment in both groups or between the two groups ($p = 0.45$, $p = 0.65$).

3.4. Complications

No complications or side effects of the treatments were reported during the study.

4. Discussion

In this study the athletes (50%) who were successfully treated with the MMT were able to return to sporting activities more quickly (12.8 weeks) when compared to those treated with the ET (55%) program (17.3 weeks). There was no significant difference in the number of athletes able to return to sporting activities between the two treatment groups. The objective treatment outcome and pain scores during sporting activities did not differ between the two treatment groups.

The results of the MMT in this study are worse than those published in a previous retrospective case series examining the effectiveness of this treatment (Weir et al., 2008). This difference may well be due to the differences in study design (prospective randomised controlled vs. case series) as the treating physicians and the work setting are the same in both studies. The athletes in this study were also involved in sporting activities at a higher level (Tegner score 8.7 (SD 0.5) vs. 7.3 (SD 1.97)) before injury and were older (28.7 vs. 20.5 years) than those in the case series which may have affected outcome.

The outcome results in the ET group are less positive than those previously published by Hölmich et al. (1999) in a randomised controlled trial. In the study of Hölmich et al. (1999) which used the same objective outcome measures, 79% of the athletes who completed the ET returned to full sporting activities without groin pain. In this study only 55% (12/22) of the athletes treated with ET had a good or excellent outcome and a full return to the pre-injury activity level. This difference may be due to the effect of supervision. In the study by Hölmich et al. (1999) the ET was performed under supervision of a physical therapist although compliance was not reported. In this study the athletes were instructed on how to perform the ET by a physical therapist on three separate occasions and the technique on performing the exercises was checked at these appointments. The athletes were however not supervised while performing the ET. In a recent randomised trial on the effects of an exercise programme for lateral elbow tendinopathy the group with supervision had significantly larger improvements than the group performing the same exercises without supervision (Stasinopoulos et al., 2009). The athletes in this study who had successful outcomes had a comparable time to return to sporting activities as those published by Hölmich et al. (1999) (17.3 vs. 18.5 weeks). These times are longer than times published in some case series although these times (4(Rodriguez et al., 2001)–10 weeks (Wollin and Lovell, 2006)) should be interpreted with caution due to the methodological considerations of the study designs.

The exact mechanism of the MMT is unknown. The program comprises different modalities and it is impossible to ascertain which components contribute to the treatment effect. The addition of a third study group in the trial comprising only certain components would have enabled more detailed analysis of the contributions of the

component parts of the MMT. A lack of this control group is a weakness of this trial and is suggested as a topic for future investigation.

It has been suggested that the MMT may work by reducing the viscoelastic muscle stiffness in the adductor muscles and thus reduce loading of the insertion and the pubic bone (Weir et al., 2008). Most athletes report a subjective feeling of reduced muscle tension directly after the Van den Akker manual therapy treatment. At present there is no generally accepted simple and reliable method to measure muscle stiffness and as such it was not measured during this study. It was also hypothesized that the MMT treatment may have affected the hip joint range of motion. The range of motion in this study (internal rotation 25°, external rotation 39°) is comparable to the range found in the recent study of Manning and Hudson (2009) who found 25° of internal rotation and 44° degrees of external rotation in asymptomatic football players of a similar age. The 25° of internal rotation found in this study is greater than the 15.5° observed in a previous study of athletes who went on to develop chronic groin injury (Verrall et al., 2007). In this study the hip joint range of motion did not alter after treatment in the MMT group and it would seem that this is not the mechanism through which the MMT may work.

It may also be that the MMT influences core stability. Core stability is often considered to play a role in athletic groin injuries. The transversus abdominus (TA) muscle has been shown to have a later onset of action in athletes with long-standing groin pain (Cowan et al., 2004). A recent structured critical review on groin pain in athletes suggested that exercises aimed at stabilizing the pelvis are important and that the TA may need some specific attention (Jansen et al., 2008a). It has been shown that a manipulation of the sacro-iliac joint can improve the feed forward action of the TA (Marshall and Murphy, 2006). A possible hypothesis of the action of the MMT is that it may affect the sacro-iliac joint and thus influence TA function. The TA function was not measured in this study as the technique requires complicated equipment and specialized training. At current there are also no other reliable simple clinical tests for core stability and thus this was not measured.

There are a number of weaknesses of this study that should be taken into account. As the treatments were physically different it was impossible to blind athletes to the treatment which may have affected outcome. The two treatment protocols are profoundly different with the MMT requiring less time and effort from the athlete than the ET. The return to running program was started four weeks earlier in the MMT group which is a structural difference that may well explain the difference in time to return to sports.

As mentioned above the lack of a control group examining different components of the MMT makes it impossible to ascertain which part of the program may help in the treatment and is a weakness. Another weakness is that in this study both of the treatments were given in isolation while in clinical practice often many treatments are combined. These weaknesses should be taken into account when interpreting the study results. The study has a relatively short follow up period of 16 weeks. This makes it difficult to assess whether or not there were recurrences of the adductor pain after completion of the treatment. The addition of an extra follow up at 24 weeks was used in cases where patients did not recover at 16 weeks. The fact that not all patients were followed up at 24 weeks is a weakness. A longer standardised follow up period would have been preferable and allowed for an assessment of recurrence rate. In future studies longer follow up periods should be used to assess for long term results and recurrence rates.

5. Conclusion

This single blinded randomised controlled clinical trial showed that the MMT was safe and equally effective treatment for athletes

with long-standing adductor-related groin pain as an ET program. After 4 months both groups showed significant decrease of VAS pain scores during sport. The athletes in the MMT group who made a full return to sports did so significantly quicker than those who performed ET. Further studies should evaluate if a combination of all treatments would improve effectiveness and reduce the time to return to sporting activities and should include a control group.

Acknowledgements

Many thanks to Aad Scheffer for making the balance boards for the exercise therapy group. Many thanks to Ton Kraan and Marcel Thomas for instructing the athletes in the exercise therapy group. Many thanks to Cora Van der Heijden, Ank de Vries and Pauline Verschoor for coordinating athletes through the study.

References

- Arnason A, Sigurdsson SB, Gudmundsson A, Holme I, Engebretsen L, Bahr R. Risk factors for injuries in football. *Am J Sports Med* 2004;32(1 Suppl):5S–16S.
- Boone DC, Azen SP, Lin CM, Spence C, Baron C, Lee L. Reliability of goniometric measurements. *Phys Ther* 1978;58(11):1355–90.
- Bovens AM, van Baak MA, Vrencken JG, Wijnen JA, Verstappen FT. Variability and reliability of joint measurements. *Am J Sports Med* 1990;18(1):58–63.
- Bradshaw C, McCrory P. Obturator nerve entrapment. *Clin J Sport Med* 1997;7(3):217–9.
- Cowan SM, Schache AG, Brukner P, Bennell KL, Hodges PW, Coburn P, et al. Delayed onset of transversus abdominus in long-standing groin pain. *Med Sci Sports Exerc* 2004;36(12):2040–5.
- Hawkins RD, Hulse MA, Wilkinson C, Hodson A, Gibson M. The association football medical research programme: an audit of injuries in professional football. *Br J Sports Med* 2001;35(1):43–7.
- Hogan AA. Rehabilitation program for osteitis pubis in football. Adelaide: OP Publications; 2006.
- Hölmich P. Long-standing groin pain in sportspeople falls into three primary patterns, a “clinical entity” approach: a prospective study of 207 patients. *Br J Sports Med* 2007;41(4):247–52.
- Hölmich P, Hölmich LR, Bjerg AM. Clinical examination of athletes with groin pain: an intraobserver and interobserver reliability study. *Br J Sports Med* 2004;38(4):446–51.
- Hölmich P, Uhrskou P, Ulnits L, Kanstrup IL, Nielsen MB, Bjerg AM, et al. Effectiveness of active physical training as treatment for long-standing adductor-related groin pain in athletes: randomised trial. *Lancet* 1999;353(9151):439–43.
- Jansen JA, Mens JM, Backx FJ, Kolfchoten N, Stam HJ. Treatment of longstanding groin pain in athletes: a systematic review. *Scand J Med Sci Sports* 2008a;18(3):263–74.
- Jansen JA, Mens JM, Backx FJ, Stam HJ. Diagnostics in athletes with long-standing groin pain. *Scand J Med Sci Sports* 2008b;18(6):679–90.
- Machotka Z, Kumar S, Perraton LG. A systematic review of the literature on the effectiveness of exercise therapy for groin pain in athletes. *Sports Med Arthrosc Rehabil Ther Technol* 2009;1(1):5.
- Manning C, Hudson Z. Comparison of hip joint range of motion in professional youth and senior team footballers with age-matched controls: an indication of early degenerative change? *Phys Ther Sport* 2009;10(1):25–9.
- Marshall P, Murphy B. The effect of sacroiliac joint manipulation on feed-forward activation times of the deep abdominal musculature. *J Manipulative Physiol Ther* 2006;29(3):196–202.
- Rodriguez C, Miguel A, Lima H, Heinrichs K. Osteitis pubis syndrome in the professional soccer athlete: a case report. *J Athl Train* 2001;36:437–40.
- Stasinopoulos D, Stasinopoulou K, Stasinopoulos I, Manias P. Comparison of effects of a home exercise programme and a supervised exercise programme for the management of lateral elbow tendinopathy. *Br J Sports Med*; Nov 3 2009 [Epub ahead of print].
- Tegner Y, Lysholm J. Rating systems in the evaluation of knee ligament injuries. *Clin Orthop Relat Res* 1985 Sep;198:43–9.
- Verrall GM, Slavotinek JP, Barnes PG, Esterman A, Oakeshott RD, Spriggins AJ. Hip joint range of motion restriction precedes athletic chronic groin injury. *J Sci Med Sport* 2007;10(6):463–6.
- Weir A, Veger SA, Van de Sande HB, Bakker EW, de Jonge S, Tol JL. A manual therapy technique for chronic adductor-related groin pain in athletes: a case series. *Scand J Med Sci Sports*; Aug 5 2008 [Epub ahead of print] PMID: 18694435.
- Wollin M, Lovell G. Osteitis pubis in four young football players: a case series demonstrating a successful rehabilitation. *Phys Ther Sport* 2006;7:153–60.